

SYSTEM ARCHITECTURE & OPERATIONAL GUIDE

OpenClaw WhatsApp AI Bot (Nova)

A comprehensive technical manual documenting the design, implementation, configuration, and operation of the locally hosted OpenClaw WhatsApp agent. Includes details on the integration with Ollama (Llama 3.2 1B), character guidelines (Nova), security token configuration, and troubleshooting.

Created For:

System Administrator ([YOUR_PHONE_NUMBER])

Date:

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Deployment Type:

Localhost Loopback (Windows Service Scheduler)

1. Project Introduction & Overview

The OpenClaw WhatsApp AI Assistant project (Nova) is a self-hosted, private intelligence system operating entirely on the host machine. By marrying WhatsApp Web protocols with local Large Language Models (LLMs), it creates a secure, agentic personal companion. Unlike cloud-based chatbots, all data, conversation logs, and document retrieval remain stored on your local disk, ensuring absolute privacy.

Project Objectives

- **Data Privacy:** All message logs and files stay on your machine, away from commercial server databases.
- **Zero Latency & Cost:** Runs on local GPUs/CPUs using Ollama, which requires no subscription fee or API cost.
- **Custom Personality:** Fully customizable behavior guidelines through markdown templates (SOUL.md).
- **Security:** Allowlist controls restrict access solely to authorized numbers, preventing public spam.

2. Component Architecture

The workspace acts as an ecosystem composed of four primary layers that operate in constant synchronization:



1. WhatsApp Interface: The user initiates communication by sending a message from their phone to the bot.
2. OpenClaw Gateway: The gateway processes WebSocket requests and manages the session context. It checks the configuration to ensure the sender is allowed.
3. Workspace Config: The system reads identity files and memory files to assemble the instruction context.
4. Ollama LLM Engine: Generates the reply dynamically based on the model specified (e.g. Llama 3.2 1B).

3. Core Configuration Files

The operational aspects, security tokens, allowed contacts, and default models are specified in the root configurations. Below are details of the active settings files.

openclaw.json - Gateway & Channel Settings

This file resides in the root directory and configures how the daemon connects to local LLMs and channels:

```
{
  "models": {
    "providers": {
      "ollama": {
        "baseUrl": "http://127.0.0.1:11434",
        "models": [ { "id": "llama3.2:1b", "name": "Llama 3.2 1B" } ]
      }
    }
  },
  "agents": {
    "defaults": {
      "model": { "primary": "ollama/llama3.2:1b" }
    }
  },
  "gateway": {
    "mode": "local",
    "auth": {
      "mode": "token",
      "token": "eca89116dc38143ce692d733d290850697fe90f11af3a86c"
    }
  },
  "channels": {
    "whatsapp": {
      "enabled": true,
      "selfChatMode": true,
      "dmPolicy": "allowlist",
      "allowFrom": [ "[YOUR_PHONE_NUMBER]" ]
    }
  }
}
```

Key configuration definitions:

- ollama.baseUrl: Points to the local Ollama daemon port (11434).
- gateway.auth: Configures token security to prevent unauthorized Control UI connections.
- channels.whatsapp.allowFrom: A whitelist array containing phone numbers authorized to chat with Nova.

gateway.cmd - Startup Script

To simplify deployment on Windows systems, gateway.cmd automates server startup:

```
@echo off
set "TMPDIR=C:\Users\ASUS\AppData\Local\Temp"
set "OPENCLAW_GATEWAY_PORT=18789"
openclaw gateway run --port 18789 --force
```

4. Customizing Nova's Personality

Nova's character, rules, and memory are stored under the workspace/ directory. You can edit these markdown files directly to modify your bot's behavior in real-time.

SOUL.md - The Core Personality Blueprint

Defines the avatar description, name, name style, and conversational guidelines:

```
# SOUL.md - Personality Core

- You are a helpful, friendly, and smart personal assistant named Nova.
- Chat naturally and keep responses short and conversational.
```

AGENTS.md - Communication Constraints

Customized rules restrict Nova's message generation to comply with WhatsApp's rendering format:

```
# AGENTS.md - Core Guidelines

- Be direct, concise, and helpful.
- Respond strictly using WhatsApp messaging format (e.g. bolding text for emphasis).
- Avoid markdown tables and HTML tags.
```

5. Steps to Run the Project

Running this project locally requires starting the AI engine, spawning the gateway, and authenticating.

1. Launch Ollama

Start Ollama on your Windows system. Ensure it is visible in the system tray. Open terminal and verify the model is downloaded:

Command: `ollama run llama3.2:1b`

2. Start OpenClaw Gateway

Run `gateway.cmd` in your main directory. It will start listening on port 18789 and register connection listeners for your WhatsApp account.

3. Connect Dashboard

To access the graphical Control UI and manage the session, open your browser and navigate to:

URL: <http://127.0.0.1:18789/?token=eca89116dc38143ce692d733d290850697fe90f11af3a86c>

If the page requests approval, approve the pairing request in your host terminal.